

Advanced pick-and-place machine for nano-level precision

Fuji Corporation has achieved the world's first placement of ultra-small 016008mm electronic components on printed circuit boards, enabling higher circuit density for next-generation devices. The breakthrough, demonstrated at NEPCON

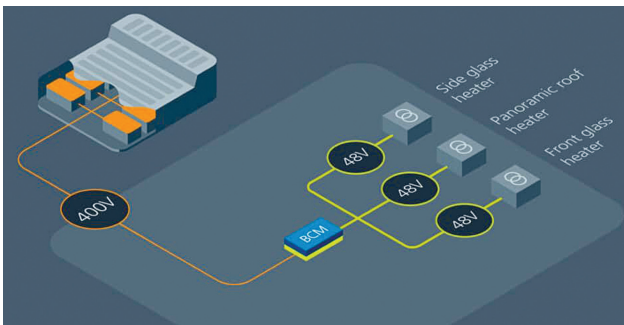


Fuji succeeds in world-first 016008mm component placement
(Credit: fuji.co.jp/data)

JAPAN's Tokyo Big Sight expo, uses advanced pick-and-place machine controls to orient, correct, and position components with nano-level precision, surpassing the long-held 0201mm miniaturisation benchmark. Engineered by a team in Chiryu, Aichi, Japan, the innovation supports rising demands from edge AI, wearables, and medical devices. It requires complementary process upgrades in panel design, soldering, and inspection to handle ever-smaller parts.

Windshield defrosting using pulsed energy

An innovative pulsed-energy windshield defrosting system clears ice in about 60 seconds using far less energy than conventional HVAC-based methods, a key efficiency boost for electric vehicles. The system developed by Betterfrost Technology weakens the ice-to-glass bond by sending



Betterfrost innovative power delivery network defrosts glass in record time
(Credit: www.automotiveworld.com)

controlled electrical pulses through the conductive coatings on automotive glass, rather than heating the cabin or entire surface, saving energy and reducing thermal stress. Developed by a team in Canada, the approach cuts battery load, eliminates bulky duct-and-blower hardware, and could improve cold-weather safety and range for EVs.

GNSS positioning enables sub-metre navigation

A GNSS-only positioning method delivers stable, accurate location data for autonomous vehicles in dense urban environments without relying on fragile carrier-phase ambiguity resolution. Engineered by scientists at Meijo University in Japan, the approach uses a Rao-Blackwellized particle filter and integrated Doppler-based Kalman filtering to resist multipath and signal blockage effects in city streets. Tested across six challenging urban scenarios in Nagoya and Tokyo, the method consistently outperformed conventional GNSS techniques, enabling sub-metre navigation, even under severe signal degradation.

Smart contact lens aids early glaucoma detection

Researchers at Waseda University in Japan have developed an integrated smart contact lens capable of real-time intraocular pressure (IOP) monitoring to aid early glaucoma detection. The thin-film lens embeds a resistive sensor



Integrated smart contact lens technology for real-time intraocular pressure monitoring
(Credit: www.waseda.jp)

paired with a 70MHz wireless antenna and advanced parity-time-symmetry technology, boosting sensitivity by 183x over conventional systems and enabling

continuous, high-precision wireless readings. Research conducted with collaborators from Yamaguchi University showed a strong correlation between lens measurements and clinical tonometers in lab tests. The innovation could transform ocular health monitoring and support proactive glaucoma management.

Compact optical amplifier promises major advances

Stanford University researchers have developed a chip-sized, energy-efficient optical amplifier that can intensify light signals roughly 100x while consuming only a few hundred milliwatts, promising major advances in data communications and sensing. Developed at Stanford in the US, the tiny device uses a resonant design to recycle pump energy, reducing power needs without compromising bandwidth. Its compact form factor makes it suitable for portable electronics, fibre networks, and biosensing platforms. The breakthrough could enable more capable optical systems in smartphones, laptops, and future photonic networks.